

# CHUAN MENG

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## RESEARCH INTERESTS

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My research lies at the intersection of Agentic AI, Information Retrieval (IR), Natural Language Processing (NLP), with a focus on **cost-effective deep research agents**.

## WORK EXPERIENCE

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**The University of Edinburgh, United Kingdom** *September 2025 - present*  
Postdoctoral Researcher (Research Associate)  
Funded by Turing AI Fellowships  
Supervisors: Dr. [Jeff Dalton](#) and Prof. [Mirella Lapata](#)

**Amazon, United Kingdom** *August 2024 - January 2025*  
Applied Scientist Intern, working on proactive conversational agents  
Manager: Dr. [Gabriella Kazai](#). Mentor: Dr. [Francesco Tonolini](#).

## EDUCATION

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**University of Amsterdam, The Netherlands** *October 2021 - June 2025*  
Ph.D. in Artificial Intelligence  
Supervisors: Prof. dr. [Maarten de Rijke](#) (ACM Fellow) and dr. [Mohammad Aliannejadi](#)  
PhD thesis: Optimizing Agentic Workflows for Information Access [\[pdf\]](#)

**Shandong University, China** *September 2018 - June 2021*  
Master's degree in Computer Science and Technology  
Supervisors: Prof. dr. [Zhumin Chen](#), dr. [Pengjie Ren](#), and dr. [Zhaochun Ren](#)  
Master's thesis: Research on Knowledge-Grounded Non-Task-Oriented Conversational System [\[pdf\]](#)

**Shandong Normal University, China** *September 2014 - June 2018*  
Bachelor's degree in Electronic Commerce

## PUBLICATIONS

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As of August 2026, I have 800+ citations (Google Scholar) with an h-index of 17. I have authored papers published in proceedings/journals, such as ACL, EMNLP, NAACL, SIGIR, CIKM, AAAI, ECIR, and TOIS.

The 4 selected top publications in the last 5 years are marked with ★

- ★ **Revisiting Text Ranking in Deep Research**  
**Chuan Meng**, Litu Ou, Sean MacAvaney, and Jeff Dalton  
**SIGIR 2026**: The 49th International ACM SIGIR Conference on Research and Development in Information Retrieval  
[\[pdf\]](#), [code](#)
- Re-Rankers as Relevance Judges**  
**Chuan Meng**, Jiqun Liu, Mohammad Aliannejadi, Fengran Mo, Jeff Dalton, and Maarten de Rijke  
Under review  
[\[pdf\]](#), [code](#)
- Toward Exploring Mixed-Initiative Conversation Generation Based on Community Question Answering**  
Lili Lu, Pranav Kasela, Federico Ravenda, **Chuan Meng**, Gabriella Pasi, and Fabio Crestani

**TOIS**: ACM Transactions on Information Systems, 44(5), 1–29, 2026

[\[pdf\]](#)

4. **Generative Retrieval with Few-shot Indexing**  
Arian Askari\*, **Chuan Meng**\* (co-first author), Mohammad Aliannejadi, Zhaochun Ren, Evangelos Kanoulas, and Suzan Verberne  
**ECIR 2026**: The 48th European Conference on Information Retrieval  
[\[pdf\]](#)
5. **Towards Adaptive Personalized Conversational Information Retrieval**  
Fengran Mo, Yuchen Hui, Yuxing Tian, Zhaoxuan Tan, **Chuan Meng**, Zhan Su, Kaiyu Huang, Jian-Yun Nie  
**CIKM 2025**: The 34th ACM International conference on Information and Knowledge Management  
[\[pdf\]](#)
6. **UniConv: Unifying Retrieval and Response Generation for Large Language Models in Conversations**  
Fengran Mo, Yifan Gao, **Chuan Meng**, Xin Liu, Zhuofeng Wu, Kelong Mao, Zhengyang Wang, Pei Chen, Zheng Li, Xian Li, Bing Yin, Meng Jiang  
**ACL 2025**: The 63rd Annual Meeting of the Association for Computational Linguistics  
[\[pdf\]](#)
7. ★ **Bridging the Gap: From Ad-hoc to Proactive Search in Conversations**  
**Chuan Meng**, Francesco Tonolini, Fengran Mo, Nikolaos Aletras, Emine Yilmaz, and Gabriella Kazai  
**SIGIR 2025**: The 48th International ACM SIGIR Conference on Research and Development in Information Retrieval  
[\[pdf, code\]](#)
8. ★ **Query Performance Prediction using Relevance Judgments Generated by Large Language Models**  
**Chuan Meng**, Negar Arabzadeh, Arian Askari, Mohammad Aliannejadi, and Maarten de Rijke  
**TOIS**: ACM Transactions on Information Systems, 43(4), 1–35, 2025  
[\[pdf, code\]](#)
9. **SOLID: Self-seeding and Multi-intent Self-instructing LLMs for Generating Intent-aware Information-Seeking Dialogs**  
Arian Askari, Roxana Petcu, **Chuan Meng**, Mohammad Aliannejadi, Amin Abolghasemi, Evangelos Kanoulas, and Suzan Verberne.  
**NAACL 2025**: The 2025 Annual Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics  
[\[pdf, code\]](#)
10. **Zero-Shot and Efficient Clarification Need Prediction in Conversational Search**  
Lili Lu, **Chuan Meng**, Federico Ravenda, Mohammad Aliannejadi and Fabio Crestani  
**ECIR 2025**: The 47th European Conference on Information Retrieval  
[\[pdf, code\]](#)
11. **Improving the Re-Usability of Conversational Search Test Collections**  
Zahra Abbasiantaeb, **Chuan Meng**, Leif Azzopardi and Mohammad Aliannejadi  
**ECIR 2025**: The 47th European Conference on Information Retrieval  
[\[pdf, code\]](#)
12. **Can We Use Large Language Models to Fill Relevance Judgment Holes?**  
Zahra Abbasiantaeb, **Chuan Meng**, Leif Azzopardi, Mohammad Aliannejadi  
**LLM4Eval**: The First Workshop on Large Language Models (LLMs) for Evaluation in Information Retrieval  
[\[pdf, code\]](#)
13. ★ **Ranked List Truncation for Large Language Model-based Re-Ranking**  
**Chuan Meng**, Negar Arabzadeh, Arian Askari, Mohammad Aliannejadi, and Maarten de Rijke.  
**SIGIR 2024**: The 47th International ACM SIGIR Conference on Research and Development in

14. **Query Performance Prediction for Conversational Search and Beyond**  
**Chuan Meng**  
**SIGIR 2024:** The 47th International ACM SIGIR Conference on Research and Development in Information Retrieval  
[\[pdf\]](#)
15. **LLM-based Retrieval and Generation Pipelines for TREC Interactive Knowledge Assistance Track (iKAT) 2023.**  
Zahra Abbasiantaeb, **Chuan Meng**, David Rau, Antonis Krasakis, Hossein A. Rahmani, and Mohammad Aliannejadi.  
**TREC 2023:** The Thirty-Second Text REtrieval Conference (**Our submitted runs ranked 1st**)  
[\[pdf\]](#)
16. **Expand, Highlight, Generate: RL-driven Document Generation for Passage Reranking**  
Arian Askari, Mohammad Aliannejadi, **Chuan Meng**, Evangelos Kanoulas, and Suzan Verberne  
**EMNLP 2023:** The 2023 Conference on Empirical Methods in Natural Language Processing  
[\[pdf\]](#), [code](#)
17. **System Initiative Prediction for Multi-turn Conversational Information Seeking**  
**Chuan Meng**, Mohammad Aliannejadi, and Maarten de Rijke  
**CIKM 2023:** The 32nd ACM International Conference on Information and Knowledge Management  
[\[pdf\]](#), [code](#)
18. **Query Performance Prediction: From Ad-hoc to Conversational Search**  
**Chuan Meng**, Negar Arabzadeh, Mohammad Aliannejadi, and Maarten de Rijke  
**SIGIR 2023:** The 46th International ACM SIGIR Conference on Research and Development in Information Retrieval  
[\[pdf\]](#), [code](#)
19. **Performance Prediction for Conversational Search Using Perplexities of Query Rewrites**  
**Chuan Meng**, Mohammad Aliannejadi, and Maarten de Rijke  
**QPP++ 2023:** Query Performance Prediction and Its Evaluation in New Tasks Workshop co-located with The 45th European Conference on Information Retrieval  
[\[pdf\]](#), [code](#)
20. **Initiative-Aware Self-Supervised Learning for Knowledge-Grounded Conversations**  
**Chuan Meng**, Pengjie Ren, Zhumin Chen, Zhaochun Ren, Tengxiao Xi, and Maarten de Rijke  
**SIGIR 2021:** The 44th International ACM SIGIR Conference on Research and Development in Information Retrieval  
[\[pdf\]](#)
21. **Conversations Powered by Cross-Lingual Knowledge**  
Weiwei Sun\*, **Chuan Meng\*** (co-first author), Qi Meng, Zhaochun Ren, Pengjie Ren, Zhumin Chen, and Maarten de Rijke  
**SIGIR 2021:** The 44th International ACM SIGIR Conference on Research and Development in Information Retrieval  
[\[pdf\]](#), [code](#)
22. **DukeNet: A Dual Knowledge Interaction Network for Knowledge-Grounded Conversation**  
**Chuan Meng**, Pengjie Ren, Zhumin Chen, Weiwei Sun, Zhaochun Ren, Zhaopeng Tu, and Maarten de Rijke  
**SIGIR 2020:** The 43rd International ACM SIGIR Conference on Research and Development in Information Retrieval  
[\[pdf\]](#), [code](#)
23. **RefNet: A Reference-aware Network for Background Based Conversation**  
**Chuan Meng**, Pengjie Ren, Zhumin Chen, Christof Monz, Jun Ma, and Maarten de Rijke  
**AAAI 2020:** The Thirty-Fourth AAAI Conference on Artificial Intelligence.

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## ACADEMIC SERVICE

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So far, I have served on the program committees of 12 major venues, and as a reviewer for 3 journals.

- Workshop organization:
  - **QPP++2025: Query Performance Prediction and its Applications in the Era of Large Language Models**  
**Chuan Meng**, Guglielmo Faggioli, Mohammad Aliannejadi, Nicola Ferro, and Josiane Mothe  
**ECIR 2025**: The 47th European Conference on Information Retrieval  
6–10 April 2025, Lucca, Italy  
[\[pdf, website\]](#)
- Senior Program Committee member: CIKM 2026
- Program Committee member: ACL 2023, EMNLP 2023–2021, SIGIR 2026–2024, COLING 2025, 2022 & 2020, CIKM 2025–2022, The Web Conf 2026–2024, WSDM 2026–2022, ECIR 2026–2024, ICTIR 2025 & 2023, SIGKDD 2022, ECML/PKDD 2022–2021, AAAI 2021
- ACL Rolling Review (ARR) reviewer 2025
- Journal reviewer: Transactions on Information Systems (TOIS), Pattern Recognition, Information Processing and Management (IP&M)
- arXiv, Moderator (cs.IR), February 2026–present

## TEACHING & SUPERVISION

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- Tutorial organization: so far, I have organized 7 tutorials, as the first speaker in 3 of them and the second speaker in the other 4.
  - **Agentic Search: Foundations, Optimization, and Long-Horizons**  
**Chuan Meng**, Litu Ou, Sean MacAvaney, and Jeff Dalton  
**CIKM 2026**: The 35th ACM International Conference on Information and Knowledge Management (upcoming)  
7 November 2026, Rome, Italy
  - **Conversational Search: From Fundamentals to Frontiers in the Age of Agents**  
**Chuan Meng**, Fengran Mo, Mohammad Aliannejadi, Jeff Dalton, and Jian-Yun Nie  
**WWW 2026**: The ACM Web Conference 2026  
30 June 2026, Dubai, United Arab Emirates  
[\[pdf, slides, website\]](#)
  - **Conversational Search: Foundations, Large Language Models, and Agents**  
**Chuan Meng**, Fengran Mo, Mohammad Aliannejadi, Jeff Dalton, and Jian-Yun Nie  
**ECIR 2026**: The 48th European Conference on Information Retrieval  
29 March–2 April 2026, Delft, The Netherlands  
[\[pdf, slides, website\]](#)
  - **Conversational Search: From Fundamentals to Frontiers in the LLM Era**  
Fengran Mo, **Chuan Meng**, Mohammad Aliannejadi, and Jian-Yun Nie  
**SIGIR 2025**: The 48th International ACM SIGIR Conference on Research and Development in Information Retrieval  
13–17 July 2025, Padua, Italy  
[\[pdf, slides, website\]](#)
  - **Query Performance Prediction: Theory, Techniques and Applications**  
Negar Arabzadeh, **Chuan Meng**, Mohammad Aliannejadi, and Ebrahim Bagheri  
**WSDM 2025**: The 18th ACM International Conference on Web Search and Data Mining  
10–14 March 2025, Hanover, Germany  
[\[pdf, slides\]](#)
  - **Query Performance Prediction: Techniques and Applications in Modern Information Retrieval**

Negar Arabzadeh, **Chuan Meng**, Mohammad Aliannejadi, and Ebrahim Bagheri

**SIGIR-AP 2024**: The 2nd ACM SIGIR-AP conference

9–12 December 2024, Tokyo, Japan

[pdf, slides]

– **Query Performance Prediction: From Fundamentals to Advanced Techniques**

Negar Arabzadeh, **Chuan Meng**, Mohammad Aliannejadi, and Ebrahim Bagheri

**ECIR 2024**: The 46th European Conference on Information Retrieval

24–28 March 2024, Glasgow, UK

[pdf, slides]

• Lecturing:

- **Recommender Systems** (course lecture, about **50** students), University of Amsterdam, June 2025

Course design and delivery of the lecture “Sequential and Large Language Model-based Recommender Systems”

[slides]

- **Conversational Search: From Fundamentals to Frontiers in the LLM Era** (guest lecture, about **40** undergraduate students), University of Massachusetts Amherst, October 2025

Host: [Razieh Negin Rahimi](#)

[slides]

• Teaching assistance:

- **Information Retrieval** (about **60** students), University of Amsterdam, 2022–2023; project design and grading

- **Information Retrieval** (about **60** students), Shandong University, 2020; assignment and project design, and grading

- **Natural Language Processing** (about **70** students), Shandong University, 2019; assignment and project design, and grading

• PhD mentoring:

- [Paul Owoicho](#), The University of Edinburgh, with Dr. [Jeff Dalton](#)

- [Lili Lu](#), Università della Svizzera italiana (USI), with Prof. dr. [Fabio Crestani](#)

## INVITED TALKS

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So far, I have delivered 10 invited talks at universities and industrial research labs.

- **The Role of Information Retrieval in Deep Research**

16 August 2026, Young Scholars Forum at the 32nd China Conference on Information Retrieval (CCIR 2026)

- **The Role of Information Retrieval in Deep Research**

24 June 2026, King’s College London (KCL)

Host: [Yulan He](#), professor at King’s College London

- **Revisiting Text Ranking in Deep Research**

5 March 2026, University of Sheffield

Host: [Nikolaos Aletras](#), professor at the University of Sheffield

- **Revisiting Text Ranking in Deep Research**

3 March 2026, the Royal Society

- **Opportunities and Challenges of LLMs in Information Retrieval**

21 October 2024, University of Glasgow

Host: Iadh Ounis, professor at the University of Glasgow

[slides, post]

- **Predicting the Right Moment for System Initiative in Mixed-Initiative Conversational Search**

26 August 2024, Amazon (London)

Host: Gabriella Kazai, principal applied scientist at Amazon

[\[slides\]](#)

- **Opportunities and Challenges of LLMs in Information Retrieval**

14 August 2024, Objective, Inc.

Host: Pablo Mendes, co-founder & CEO at Objective, Inc.

[\[slides\]](#)

- **Opportunities and Challenges of LLMs in Information Retrieval**

17 April 2024, Amazon (Seattle)

Host: Shervin Malmasi, applied science manager at Amazon

[\[slides\]](#)

- **System Initiative Prediction and Query Performance Prediction for Conversational Information Seeking**

3 November 2023, University College London (UCL)

Host: Xi Wang, lecturer at the University of Sheffield

[\[slides\]](#), [\[post\]](#)

- **Query Performance Prediction for Conversational Search**

18 May 2023, University of Glasgow

Host: Iadh Ounis, professor at the University of Glasgow

[\[slides\]](#), [\[post\]](#)

## ADMINISTRATIVE ACTIVITIES

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- Manager for the [IRLab LinkedIn](#) account, 2025
- Webmaster for the [IRLab website](#), 2023–2024
- Chair for internal seminars at IRLab, 2023
- Lead organizer of the IRLab BBQ event, 2023

## RESEARCH FUNDING PROPOSALS

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I have contributed to the methodology design of 3 national research funding proposals in China.

- **National Key Research and Development Program of China** (Project No.: 2020YFB1406700)  
Contributor (methodology design)  
Development of an AI-assisted writing robot system based on news reporting scenarios
- **National Natural Science Foundation of China (NSFC)** (Project No.: 61972234)  
Contributor (methodology design)  
Research on Explainable Medical Health Question-Answering Technology Based on Multi-Source Heterogeneous Data Fusion
- **National Natural Science Foundation of China (NSFC)** (Project No.: 62072279)  
Contributor (methodology design)  
Research on Conversational Recommendation Systems and Functional Enhancements Based Primarily on Implicit Feedback Data

## SCHOLARSHIPS & AWARDS

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- Excellent Master's Thesis of Shandong Province, 2022
- National Scholarship (China), 2020/2016
- Outstanding Graduates of Shandong Province, 2021/2017
- SIGIR Student Travel Grant, 2020
- Scholarship for Outstanding Postgraduate Cadres of Shandong University, 2020/2019
- AAAI Student Scholar Scholarship, 2019
- Academic Scholarship for Master Students of Shandong University, 2019
- Outstanding Students of Shandong Province, 2017

## RESOURCES

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I have curated the following resources:

- a [resource repository](#) on Hugging Face (**over 14,000 downloads**) providing the passage corpus, pre-built retriever indexes, and agent execution trajectories for deep research.
- a [code repository](#) (**over 200 stars**) providing scripts for text ranking in deep research, within a Python/PyTorch framework.
- a [code repository](#) (**over 100 stars**) providing scripts for fine-tuning open-source LLMs to generate relevance judgments, within a Python/PyTorch framework.
- a [code repository](#) (**over 110 stars**) providing a comprehensive implementation of query performance prediction (QPP) methods, within a unified Python/PyTorch framework.
- a [code repository](#) (**over 50 stars**) offering a comprehensive implementation of ranked list truncation methods, within a unified Python/PyTorch framework.
- a [paper reading list](#) (**over 250 stars**) on knowledge-grounded dialogue systems.

Also, I have contributed to key open-source toolkits in information retrieval, including:

- [Pyserini \(v0.19.2\)](#), a Python toolkit for reproducible information retrieval research.
- [Tevatron](#), a flexible and efficient Python toolkit for training and inference of neural retrieval models.

## REFERENCES

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Dr. [Jeff Dalton](#)

Turing AI Acceleration Fellow

Chancellor's Fellow (Associate Professor) at The University of Edinburgh

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Prof. dr. [Maarten de Rijke](#)

ACM Fellow

Distinguished University Professor at the University of Amsterdam

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